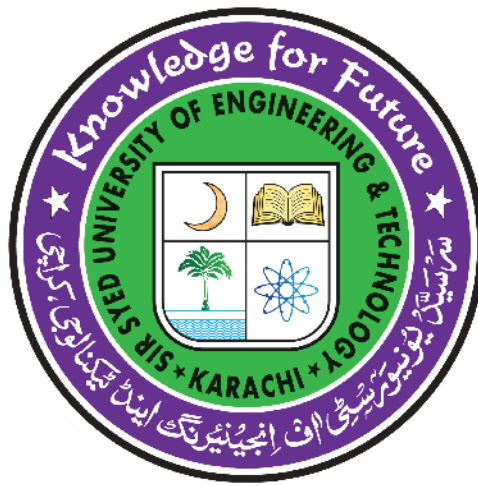


Laboratory Manual

Artificial Intelligence (CS – 325L)

6th Semester Spring 2024



Student Name: _____

Student Roll# : _____ Section : _____

BS in Computer Science
Department of Computer Science & Information Technology
Sir Syed University of Engineering and Technology
University Road Karachi - 75300
<http://www.ssuet.edu.pk>

Course Learning Outcome(s):

No.	Course Learning Outcomes (CLOs)	PLOs	Bloom's Taxonomy
1	Follow the instructions to develop working knowledge of Artificial Intelligence, Machine learning, deep learning and natural language processing algorithms in programming languages.	PLO_5 (Modern Tool Usage)	P3 (Guided Response)
2	Construct a project to implement different Artificial Intelligence algorithms in programming language.	PLO_4 (Design/ Development of Solutions)	P4 (Mechanism)
3	Demonstrate the knowledge which will be assessed through viva and presentation.	PLO_7 (Communication)	A3 (Valuing)

Laboratory Experiments

Lab #	Topics	Signature
1	Familiarization with Python language and how to use different syntax to control indexing using strings, loop and conditional statement	
2	Familiarization with Python language using functions, list, tuple and dictionary	
3	Familiarization with the concepts of class and object using Python language	
4	Manage a set of records with priority queue using queue and heapq module in python	
5	Implement shortest path using Breadth First Search (BFS) and Depth First Search (DFS) techniques in python	
6	Implement informed A* Search, Best First Search and Greedy Best First Search algorithms using Python	
7	Familiarize students with the NumPy and Matplotlib libraries of python	
8	Familiarize students with prolog programming, rules and facts. Also exploring operators and built in predicates of prolog	
9	Perform supervised learning Decision Tree and Naïve Bayes algorithms for training, testing and classification using Python	
10	Perform unsupervised learning K-means clustering algorithm for training, testing and classification using Python	
11	Implement Artificial Neural Network (ANN) and Convolutional Neural Network (CNN) using Python	
12	Implement various functions related to Natural Language Processing (NLP) using Python	
13	Getting started with OpenCV on Jupiter Lab	
14	Open Ended Lab	

Laboratory Rubrics

Criteria	Exceeds Expectations (≥90%)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory (<50%)
Software Handling (2)	Able to use software with its standard and advanced features without assistance	Able to use software with its standard and advanced features with minimal assistance	Able to use software with its standard features with assistance	Unable to use the software
Programming/ Simulation (5)	Able to program/ simulate the lab tasks with simplification	Able to program/ simulate the lab tasks without errors	Able to program/ simulate lab tasks with errors	Unable to program/ simulate
Results (2)	Able to achieve all the desired results with alternate ways to improve measurements	Able to achieve all the desired results	Able to achieve most of the desired results with errors	Unable to achieve the desired results
Lab Report (1)	Laboratory manual has no grammatical and/ or spelling errors. All sections of the report are very well written and technically accurate.	Laboratory manual has very few grammatical/ spelling errors. All sections of the report are technically accurate.	-Laboratory manual has multiple grammatical/ spelling errors. Few sections of the report contain technical error s.	Laboratory manual has several grammatical/ spelling errors and sentence construction is poor. -All sections of the report contain multiple technical errors.

Laboratory Report Score

Lab#	Criteria				Score
1.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
2.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
3.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
4.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
5.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
6.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
7.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
8.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
9.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
10.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
11.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
12.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
13.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
14.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
Obtained Score out of (140)					
Obtained Score out of (X) = Obtained Score out of (140) /140* X					

Semester Project Rubric

Criteria	Exceeds Expectations ($\geq 90\%$)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory ($< 50\%$)	Score
Project Demonstration (5)	Able to demonstrate the project with achievement of required objectives having clear understanding of project limitations and future enhancements. Hardware and/or Software modules are fully functional, if applicable.	Able to demonstrate the project with achievement of required objectives but understanding of project limitations and future enhancements is insufficient. Hardware and/or Software modules are functional, if applicable.	Able to demonstrate the project with achievement of at least 50% required objectives and insufficient understanding of project limitations and future enhancements. Hardware and/or Software modules are partially functional, if applicable.	Able to demonstrate the project with achievement of less than 50% required objectives and lacks in understanding of project limitations and future enhancements. Hardware and/or Software modules are not functional, if applicable.	
Project results (3)	Able to achieve all the desired results with alternate ways to improve measurements	Able to achieve all the desired results	Able to achieve most of the desired results with errors	Unable to achieve the desired results	
Report Writing (3)	Project report has no grammatical and/ or spelling errors. All sections of the report are very well-written and technically accurate.	Project report has very few grammatical/ spelling errors. All sections of the report are technically accurate.	Project report has multiple grammatical/ spelling errors. Few sections of the report contain technical errors.	Project report has several grammatical/ spelling errors and sentence construction is poor.	
Viva (4)	Able to answer the questions easily and correctly across the project.	Able to answer the questions related to the project	Able to answer the questions but with mistakes	Unable to answer the questions	
Obtained Score out of (15)					
Obtained Score out of (X)= Obtained Score out of (15) /15* X					

Laboratory Examination Rubric

Criteria	Exceeds Expectations ($\geq 90\%$)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory ($< 50\%$)	Score
Performance (15)	Able to present full knowledge of both problem and solution.	Able to present adequate knowledge of both problem and solution	Able to present sufficient knowledge of both problem and solution	No or very less knowledge of both problem and solution	
Viva (5)	Able to answer the questions easily and correctly across the project.	Able to answer the questions related to the project	Able to answer the questions but with mistakes	Unable to answer the questions	
Obtained Score out of (20)					
Obtained Score out of (X)= Obtained Score out of (20) /20* X					

Final Lab Assessment

Criteria	Score
Laboratory Report (15)	
Semester Project (15)	
Laboratory Examination (20)	
Total Score out of (50)	

Examiner: _____

(Name and Signature)